

Heart Stroke Prediction



Machine Learning (ML) Approach to Early Detection

Data Reference: <https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>

Github Repository: github.com/jansherkhan-data-scientist/Heart-Stroke-Prediction



Project Objectives

Risk Prediction



Build predictive models to identify patients at high risk of stroke before symptoms appear.

Key Factor Analysis



Analyze health indicators and determine which factors contribute most to stroke occurrence.

Algorithm Comparison



Compare Logistic Regression, SVM, Decision Tree & KNN across accuracy, F1 score, and loss metrics.

Early Intervention



Support healthcare providers in designing early prevention and screening strategies.

Dataset Overview



5,110

Patient Records

Source: Healthcare Stroke Dataset

Feature	Description	Type
gender	Male, Female, Other	Categorical
age	Age in years (0–82)	Numeric
hypertension	0 = No, 1 = Yes	Binary
heart_disease	0 = No, 1 = Yes	Binary
ever_married	Yes / No	Categorical
work_type	Private, Self-employed, Govt, etc.	Categorical
Residence_type	Urban / Rural	Categorical
avg_glucose_level	Average glucose level (55–272 mg/dL)	Numeric
bmi	Body Mass Index (10–97)	Numeric
smoking_status	Formerly smoked / Never / Smokes / Unknown	Categorical
stroke (Target)	0 = No Stroke, 1 = Stroke	Binary

Data Preprocessing

1

Remove ID Column

The patient ID field is a unique identifier — not relevant for pattern learning and removed from training.

2

Handle Missing Values

BMI column had 201 missing entries. Filled using Mode Imputation to maintain distribution integrity.

3

Label Encoding

Converted categorical text variables (gender, marital status, residence, work type, smoking) into numeric codes.

4

Age Binning

Ages grouped into 5 bins: Child (0–12), Teen (13–19), Young Adult (20–30), Adult (31–60), Senior (61–100).

Label Encoding Mapping

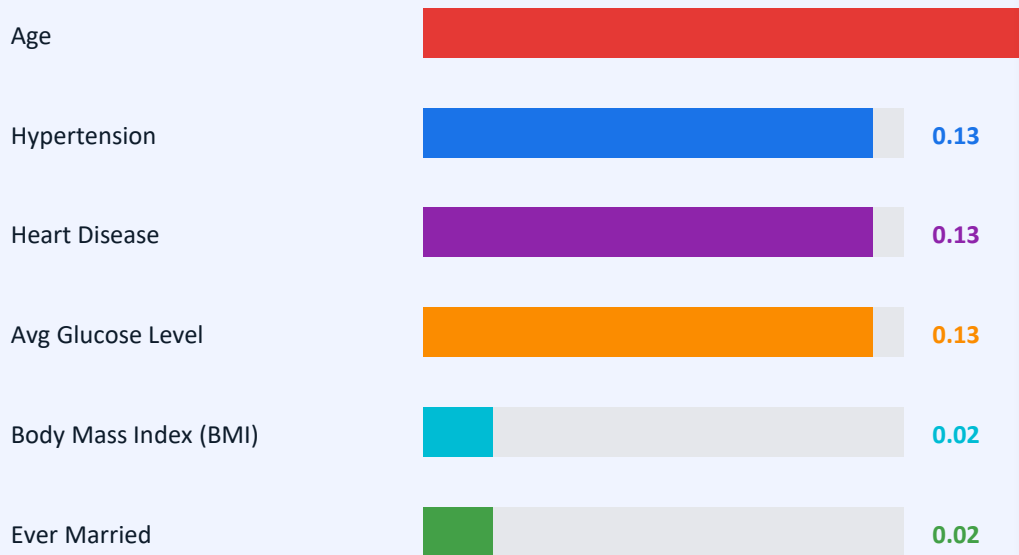
Feature	Encoding Mapping
Gender	Male = 1 Female = 0 Other = 2
Ever Married	Yes = 1 No = 0
Residence Type	Urban = 1 Rural = 0
Work Type	Private = 0 Self-employed = 1 Children = 2 Govt_job = 3 Never_worked = 4
Smoking Status	Formerly smoked = 0 Never smoked = 1 Smokes = 2 Unknown = 3

Age Binning Groups

0–12 Child	13–19 Teenager	20–30 Young Adult	31–60 Adult	61–100 Senior
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Key Insights from EDA (Exploratory Data Analysis)

Correlation with Stroke Risk



Interesting Observations



Non-smokers showed higher stroke probability than active smokers — a counter-intuitive finding.



Married individuals displayed higher stroke incidence — possibly a confounding age factor.

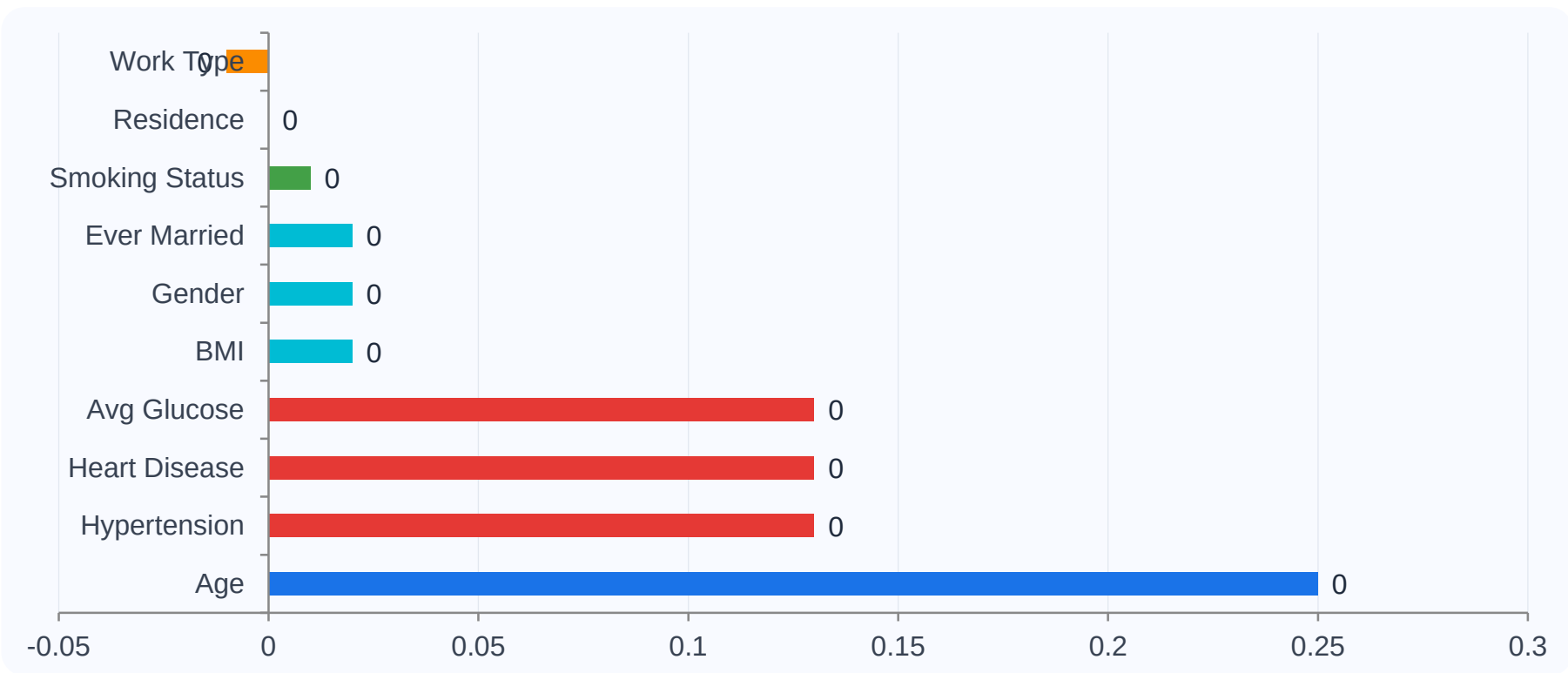


Body Mass Index (BMI) between 20–50 correlated with elevated stroke risk.



Urban residents showed marginally higher rates, likely due to lifestyle and stress.

Feature Correlation Heatmap



★ Age is the strongest predictor of stroke

Model Training Setup

Training / Test Split

80% Training — 4,088 Samples

20% Testing — 1,022



Logistic Regression (LR)

Linear model estimating probability of binary outcome using the sigmoid (logistic) function.



Support Vector Machine (SVM)

Finds the optimal hyperplane that best separates classes in high-dimensional feature space.



Decision Tree (DT)

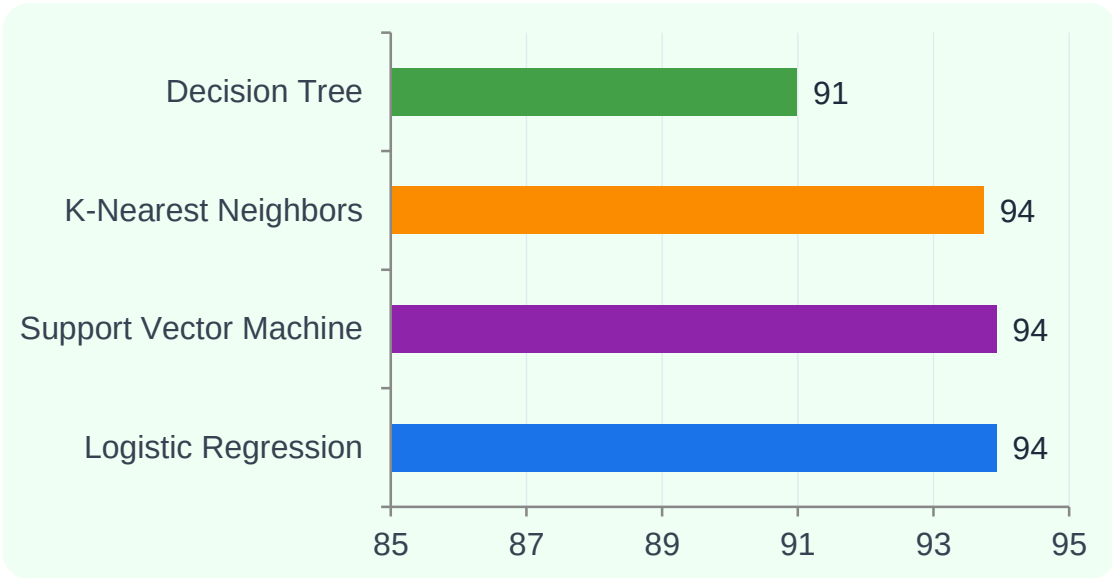
Recursive partitioning algorithm that splits data based on feature thresholds to form a tree.



K-Nearest Neighbors (KNN)

Instance-based learner that classifies a data point by majority vote of its K nearest neighbors.

Model Performance Comparison



Model	Accura cy	F1 Score	Log Loss
Logistic Regression (LR)	93.93%	0.00	2.19
Support Vector Machine (SVM)	93.93%	0.00	2.19
K-Nearest Neighbors (KNN)	93.74%	0.00	2.26
Decision Tree (DT)	90.99%	0.25	3.24

Tied for Best

Logistic Regression (LR) & SVM both achieved 93.93% accuracy.

Class Imbalance

F1 = 0.0 for most models reveals the dataset heavily favors the non-stroke class.

Best F1 Score

Decision Tree (DT) alone detected some stroke cases — F1 = 0.25 despite lower accuracy.

Note: Higher accuracy ≠ better model when data is imbalanced. F1 Score is critical for medical diagnostics.

Confusion Matrices — All Models

Logistic Regression (LR)

960

True Negatives (TN)

0

False Positives (FP)

62

False Negatives (FN)

0

True Positives (TP)

SVM

960

True Negatives (TN)

0

False Positives (FP)

62

False Negatives (FN)

0

True Positives (TP)

KNN

959

True Negatives (TN)

0

False Positives (FP)

63

False Negatives (FN)

0

True Positives (TP)

Decision Tree (DT)

965

True Negatives (TN)

5

False Positives (FP)

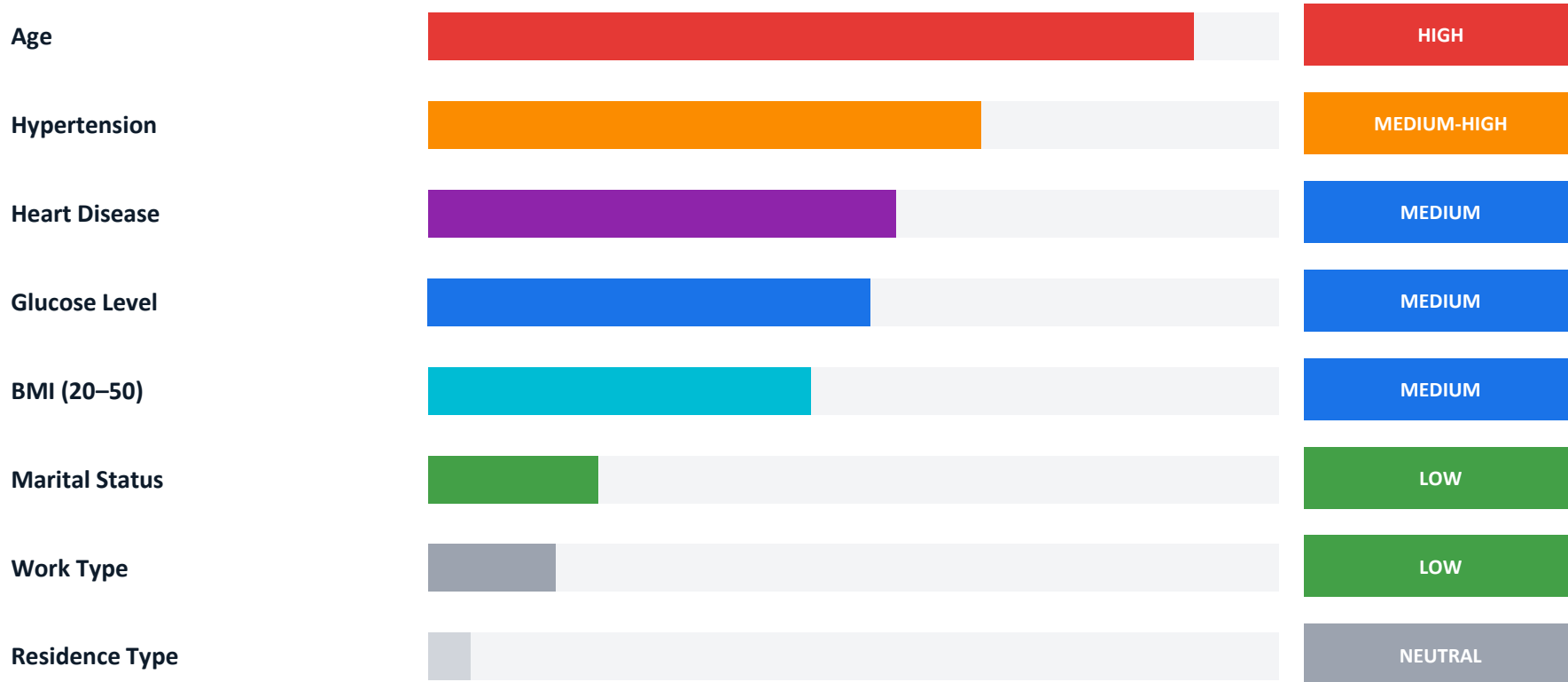
57

False Negatives (FN)

0

True Positives (TP)

Key Risk Factors Identified



Unusual & Unexpected Findings

🔍 Smoking Paradox

Counter-intuitively, non-smokers showed HIGHER stroke probability than active smokers in this dataset. This likely reflects confounding variables — many former smokers were already older or had prior illness.

💔🔍 Heart Disease Reversal

In some visualizations, patients with lower reported heart disease correlated with increased stroke incidence. Suggests possible underdiagnosis or that other factors dominate the prediction.

🔍 Hypertension Anomaly

A paradoxical pattern was observed for hypertension in certain cross-plots. May indicate interactions with medication, reporting bias, or survivorship effects in the data.

🔍 Recommendation: These findings require a larger, more balanced dataset and deeper clinical investigation before drawing clinical conclusions.

Recommendations



For Healthcare Providers

- Screen elderly patients (age 61–100) as a top priority.
- Closely monitor blood pressure (hypertension) and cardiac history.
- Do not disregard non-smokers — they still carry meaningful risk.
- Conduct regular average glucose level checks for at-risk groups.



For Model Deployment

- Use Logistic Regression (LR) or SVM as a baseline model in deployment.
- Apply SMOTE (Synthetic Minority Oversampling Technique) to balance classes.
- Explore ensemble methods (Random Forest, XGBoost) for better minority detection.
- Continuously retrain models as new patient records are collected.

Future Improvements & Next Steps

1. Address Class Imbalance



- Apply SMOTE (Synthetic Minority Oversampling Technique)
- Use class-weighted model parameters
- Collect more stroke-positive (minority class) samples

2. Advanced ML Models



- Random Forest — ensemble of Decision Trees
- XGBoost / LightGBM — gradient boosting methods
- Artificial Neural Networks (ANN) for deeper patterns

3. Feature Engineering



- Create interaction features (age $\times$ glucose, BMI $\times$ hypertension)
- Add family history, medication data, cholesterol levels
- Include longitudinal health tracking over time

Conclusion

Key Takeaways



All models achieved ~94% accuracy in predicting non-stroke (negative) cases successfully.



Class imbalance severely limits detection of actual stroke events — F1 Score must be improved.



Age is the strongest and most significant predictor of stroke risk across all models.



Hypertension and Heart Disease are major clinical risk factors confirmed by correlation analysis.



With SMOTE and advanced models, this system can effectively support early stroke screening.





Thank You!

Questions & Discussion Welcome

Python 3.11

Pandas

NumPy

Scikit-Learn

Matplotlib

Seaborn